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EDUCATION

MLR Institute of Technology, Hyderabad

Computer Science and Engineering

Aug 2023 - May 2027 (expected)

7.0 CGPA

PROJECTS

Hotel Booking Backend  | *Go, TypeScript, MySQL, Redis, BullMQ*

- Designed and implemented a microservices-based booking platform using Go and TypeScript, with an Auth Service acting as an API Gateway handling JWT authentication, reverse proxy routing, and request rate limiting.
- Implemented high-concurrency booking workflows using MySQL transactions, idempotent APIs, Redis-based distributed locking (Redlock), and BullMQ for asynchronous task processing.
- Developed event-driven background systems using cron jobs and message queues to handle email notifications and scheduled tasks reliably.
- Improved system reliability using MySQL across all services with Prisma and raw SQL queries, along with centralized logging and custom error handling.

Distributed Code Evaluation Platform  | *Node.js, TypeScript, Redis, BullMQ, Docker, MongoDB*

- Architected a **microservices-based coding platform** with API Gateway, Submission, Problem, Competition, and Evaluation services enabling scalable and decoupled request processing.
- Implemented a **distributed job processing pipeline using BullMQ and Redis**, allowing concurrent worker-based evaluation outside the request lifecycle.
- Built a **Docker-based sandbox execution engine** enforcing strict **CPU, memory, and execution time limits** to securely run untrusted user code.
- Designed a **competition workflow system** supporting timed contests, participant validation, and submission tracking integrated with the evaluation pipeline.

GPT-2 from scratch  | *Python, NumPy, Transformers, Deep Learning, PyTorch*

- Implemented a mini-GPT2 **124M** architecture from scratch using **only NumPy**, covering **multi-head self-attention, layer norm, causal masking, and token/positional embeddings**.
- Trained the model on **custom dataset** with a manually written **autoregressive loss loop**, enabling character-level text generation with no external ML libraries.
- Focused on raw low-level understanding and minimalism, achieving a functioning **GPT-like language model** with end-to-end tokenize from scratch.

Network Security  | *Python, FastAPI, MLflow, MongoDB, AWS EC2*

- Designed an end-to-end **MLOps pipeline** for phishing detection, covering data ingestion, validation, transformation, feature engineering, and model training using classical ML algorithms.
- Enabled **reproducible experimentation** through MLflow and DagsHub with experiment tracking, artifact versioning, and comparative model evaluation.
- Deployed a production-ready **FastAPI inference service** on **AWS EC2**, integrating trained models for real-time classification and scalable serving.

TECHNICAL SKILLS

Programming Languages: Go, TypeScript, Python, C++

Backend Technologies: Node.js, Express, GoFiber, FastAPI

Databases: MySQL, MongoDB

Messaging and Caching: Redis, BullMQ, RabbitMQ

Architecture and Concepts: Microservices, API Gateway, JWT Authentication, Distributed Systems, Idempotency, Rate Limiting

Machine Learning and MLOps: Machine Learning, Deep Learning, MLflow

CERTIFICATIONS

- Deep Learning & Reinforcement Learning — DeepLearning.AI (Stanford), NPTEL, Scaler

ACHIEVEMENTS

- Runner-up in Zignasa2k24 (Data Science track), evaluated on applied machine learning and real-world problem solving.